Probabilistic Reward Learning

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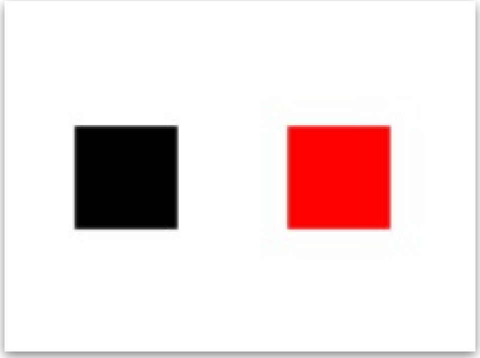
Probabilistic Reward Learning

Task Design and Conditions

11 subjects

Choose between two colors and learn which one gives more rewards based on feedback.



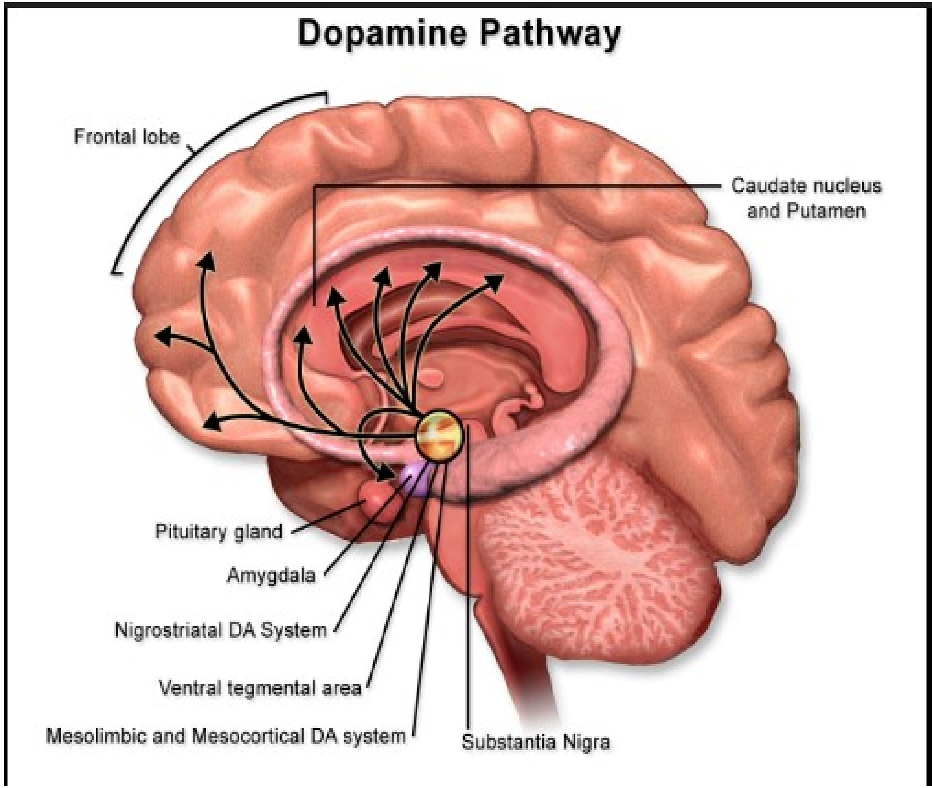
Uses in Neuroscience

Dopamine-Driven Reinforcement Learning

Learning in this task is supported by dopamine signals that come from midbrain regions like the ventral tegmental area (VTA) and substantia nigra.

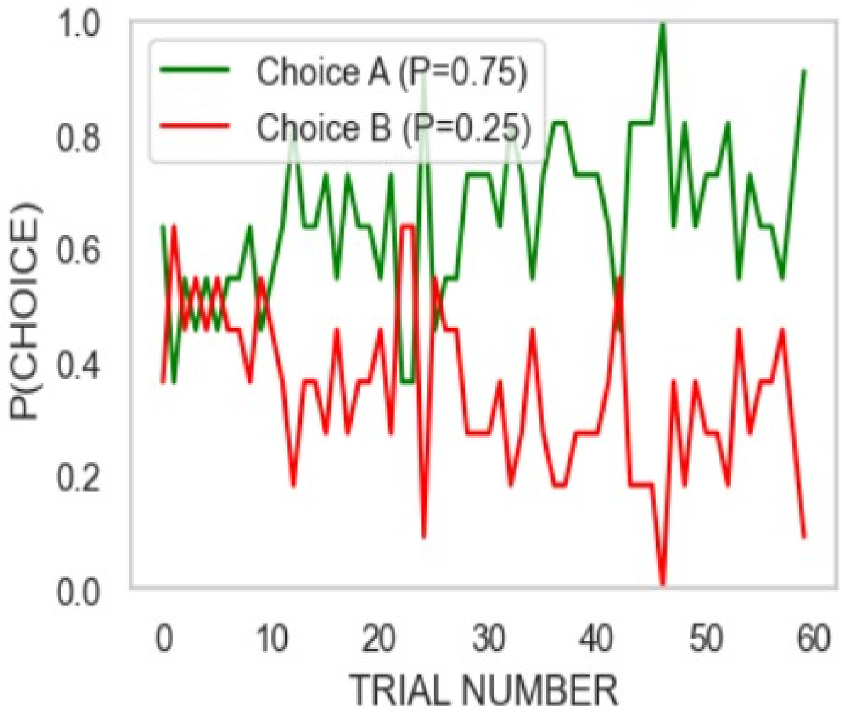
These dopamine signals travel to the striatum (caudate and putamen), helping the brain learn which choices lead to rewards.

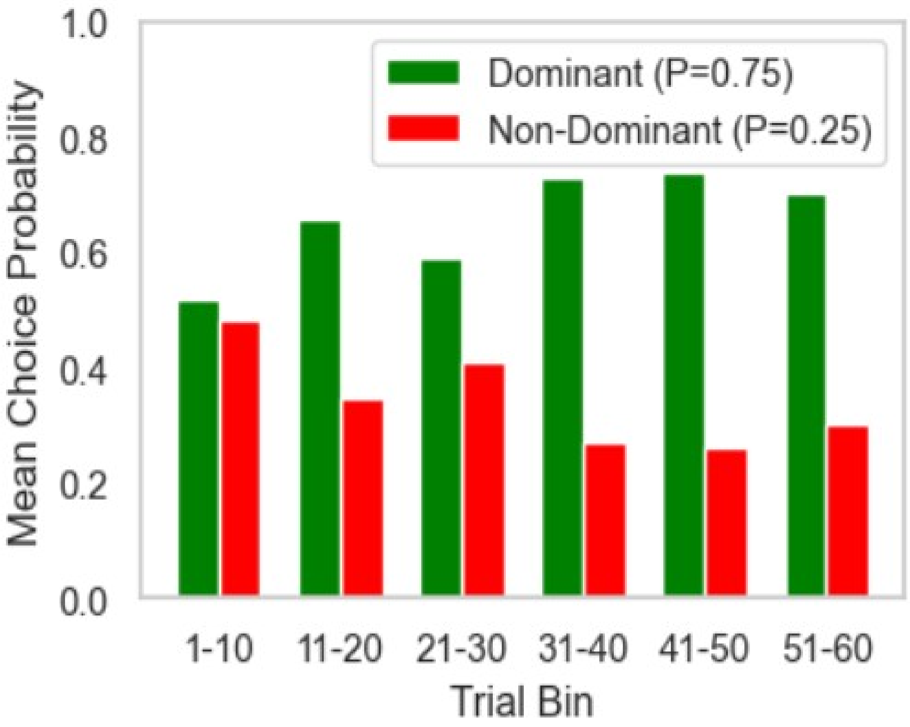
Probabilistic reward-learning tasks use this mechanism to study dopamine-based prediction-error learning, to model behavior with computational reinforcement-learning frameworks, and to understand reward sensitivity in both healthy and clinical populations.



Probabilistic Reward Learning

Results





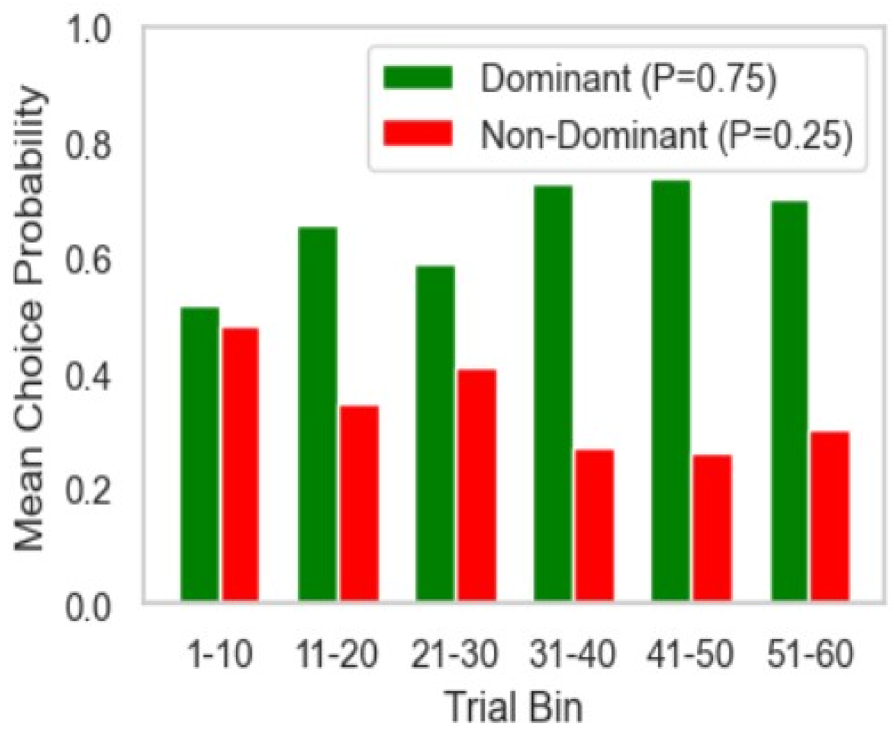
Probabilistic Reward Learning

Results

ANOVA:

No significant improvement in learning across bins over time.

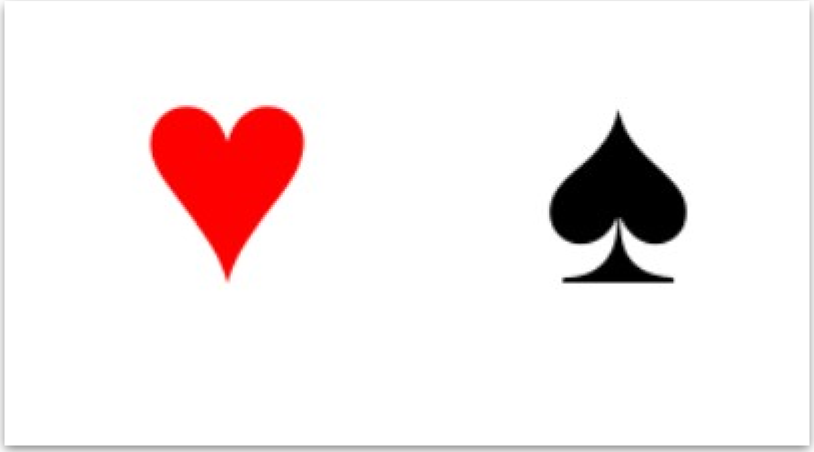
$F = 2.07$, $p = 0.08$ (>0.05)



Probabilistic Reward Learning

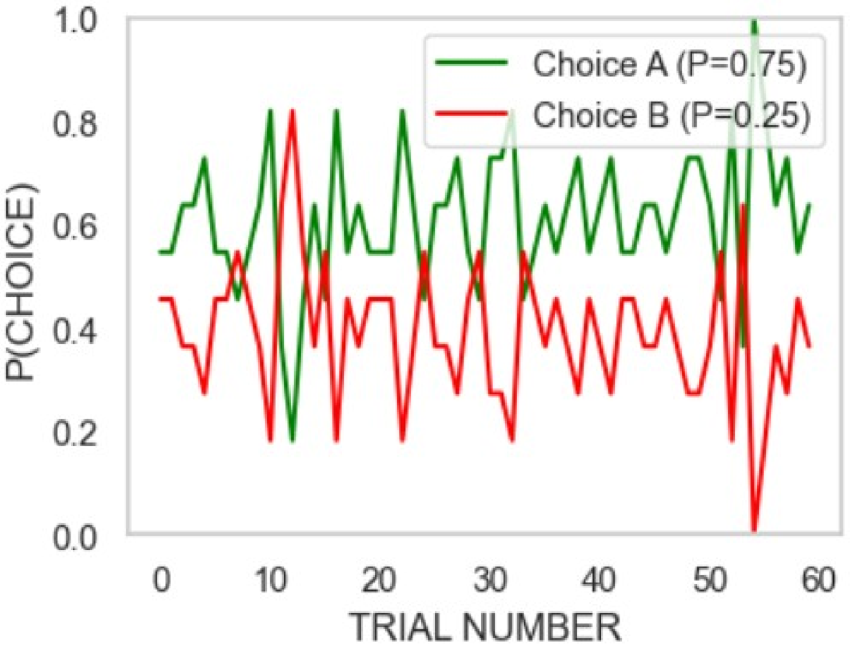
Task Design and Conditions

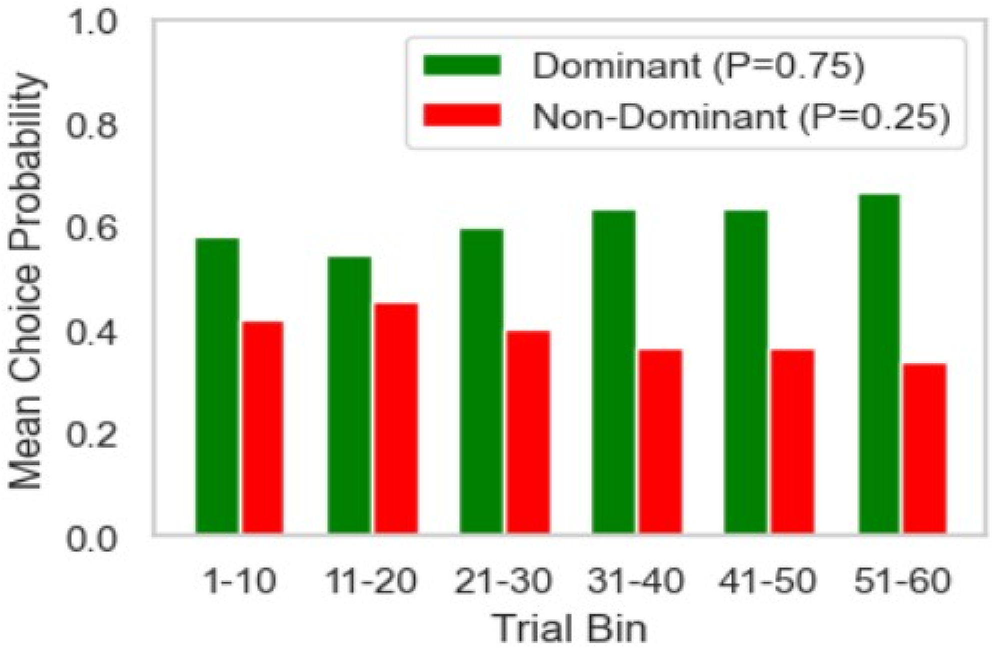
Choose between two colors and learn which one gives more rewards based on feedback. Shapes are added in this block, but they are irrelevant and act only as visual noise.



Probabilistic Reward Learning

Results





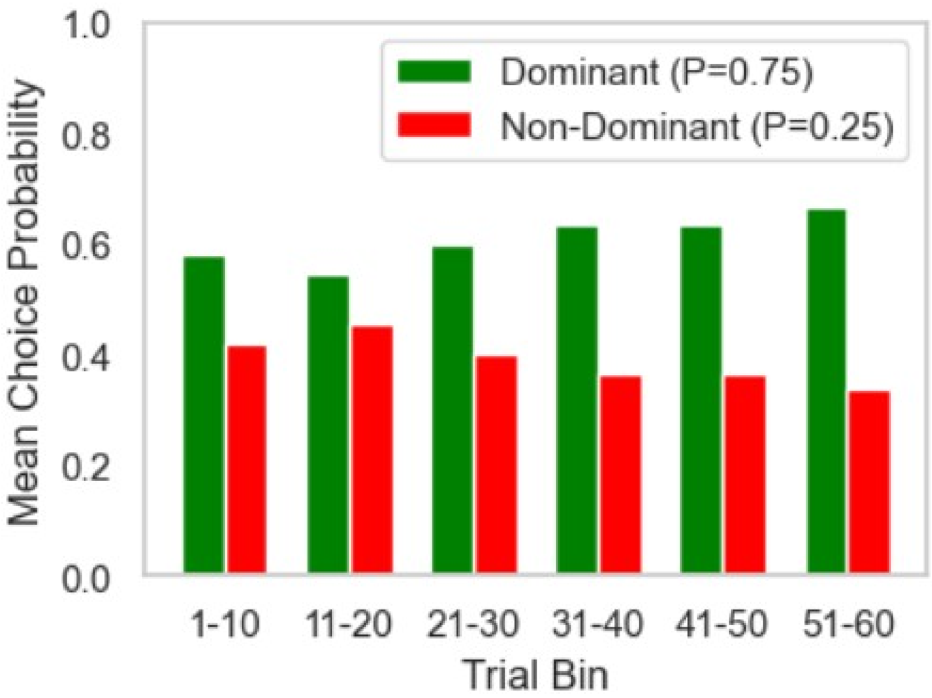
Probabilistic Reward Learning

Results

ANOVA:

No significant improvement in learning across bins over time.

$F = 5.89$, $p = 0.65$ (>0.05)



Probabilistic Reward Learning

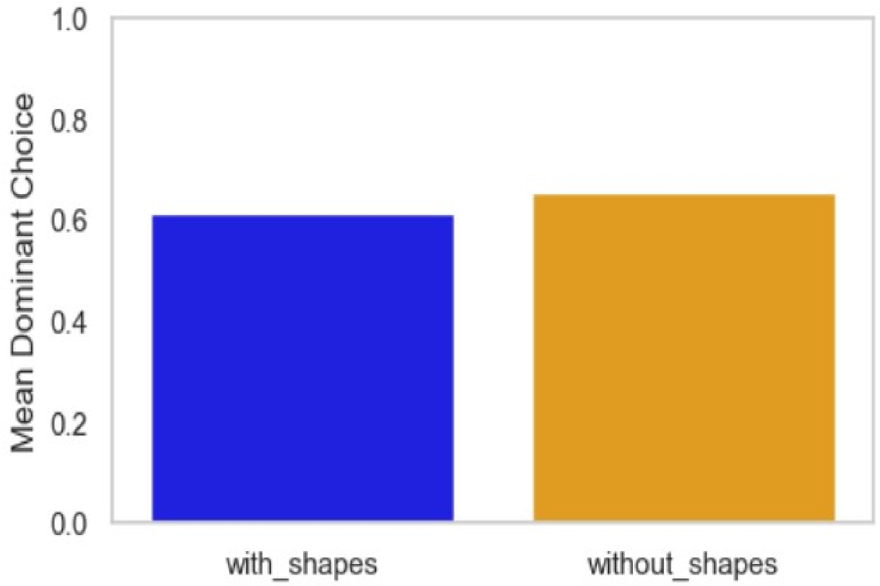
Results

Comparing learning between the with shapes and without shapes blocks.

Paired t-test:

No significant difference in learning between with shapes and without shapes.

$t(10) = -0.87$, $p = 0.40$ (>0.05)



Conclusions

While the descriptive statistics followed the expected pattern, none of the effects were statistically significant.

Participants showed a general tendency to choose the dominant (75%) option, but this improvement over time was not statistically significant in either the with-shapes or without-shapes block. Mean accuracy was slightly higher in the without-shapes block, but this difference was also not statistically significant.

One likely reason for the lack of significant findings is the small sample size (11 participants) and relatively short task length (60 trials per block). Similar probabilistic learning studies typically use 40–100 participants and over 100 trials to detect subtle learning effects. With more data, stronger and more generalizable effects might have been observed, and learning may have become more apparent.